PTPC: Prototype-Based Test-Time Prediction Calibration for Few-Shot Object Detection

Supplementary Material

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This supplement provides the per-cell evidence behind the summary numbers in the main paper. Section 3 gives protocol details for the repeated-run and support-draw studies; Section 4 derives the exact fixed-output ceiling on MS COCO; and Section 5 reports the native-host label-budget checks, cross-dataset-frozen Guard/binary-limit controls, explicitly auxiliary-label VOC/COCO diagnostics, and a controlled query-time benchmark. Section 6 contains the frozen-encoder and prototype-geometry studies; Sections 7 and 8 analyze Guard features, reference-set coverage, class-level recall risk, and parameter sensitivity; Section 9 compares simpler scoring choices; Section 10 reports detector-family transfer; and Sections 11 and 12 provide the paired failure and selectivity analyses. All results use the protocol and hyperparameters of the main paper unless stated otherwise. Unqualified "Guard" and "PTPC" denote the canonical rule in the main paper. Archived robustness and mechanism studies that use its binary limit are marked explicitly and are controls, not reruns of the canonical rule.

Experimental Protocol and Implementation Details

Data, detectors, and frozen outputs. Full PTPC is evaluated with DeFRCN and MFDC on PASCAL VOC and MS COCO under the SNIDA protocol; FSCE supplies an additional detector-family control. The encoder study covers DINOv2-S/B/L/G, CLIP-B/L, and SigLIP-B/L. Native Guard-only transfer uses FSOD-VFM and No Time to Train. DeFRCN and MFDC are rerun with the authors' code and canonical seed-0 supports. We verify that both scoring variants share identical checkpoints, support annotations, post-NMS boxes, labels, detection counts, and output order; only confidence scores change. Any upstream threshold or top-k filter is part of the unchanged host output. These seed-0 detector means need not equal published multi-run averages; published rows in the main paper are unpaired context rather than controls.

Full-PTPC configuration. The contextual novel-prototype module uses its patch-RoI score with $(w_d, w_\phi, \tau) = (.4, .6, .1)$; the crop descriptor is reserved for the Guard. The Guard uses $(m, \gamma) = (.05, .3)$, and both score bands are $(.05, 1]$. Each base prototype averages $M = 20$ annotated crops from a deterministic canonical-loader prefix, or all available crops when fewer than 20 exist. Every non-η choice comes from one VOC Split 1 development configuration and is frozen across CNN detectors, VOC Splits 2–3, and COCO. This development evidence includes labels beyond the K supports. The margin temperature is instead transferred across benchmarks: COCO source validation supplies $\eta = .04$ for every VOC cell, and VOC Split 1 supplies $\eta = .01$ for every COCO cell. DINOv2 uses $S_\phi = 518, S = 224$, patch 14; CLIP-B/16 and SigLIP-B/16 use $S_\phi = S = 224$, patch 16; CLIP-L/14 uses $S_\phi = S = 224$, patch 14; and SigLIP-L/16 uses $S_\phi = S = 256$, patch 16.

Guard-only regimes and selection safeguards. FSOD-VFM and No Time to Train reuse their mask-pooled DINOv2 descriptors. For strict cross-dataset transfer, FSOD-VFM uses the COCO-selected $(m, \gamma, (\ell_G, u_G]) = (.01, .85, (.05, 1])$, $\eta = .04$ on VOC and the VOC-Split-1-selected $(.05, .50, (.05, 1])$, $\eta = .01$ on COCO. No Time uses its own source selections: COCO supplies $(.01, .70, (0, .70])$, $\eta = .04$ for every VOC target, and VOC Split 1 supplies $(.01, .50, (0, .70])$, $\eta = .005$ for both COCO targets. The temperature grid is $\{.0025, .005, .01, .02, .04, .08\}$. Target-task labels only construct the normal support prototypes and never select these values.

The explicitly auxiliary regime constructs a deterministic 1,000-image labelled trainval population after removing all images used by the relevant base and novel supports; it is test-disjoint but exceeds the strict K-shot label budget. Each host diagnostic evaluates nine margins, six suppression factors, and 12 score bands (648 tuples). Selection ranks AP50, AP, AP75, and fire rate, reevaluates the leading 40 tuples over three deterministic folds, and rejects settings with full-set AP/AP75 losses above 0.10/0.20 point or worst-fold AP50 loss above 0.25 point. One selected tuple is then frozen across query-test shots. Selected settings, grids, support exclusions, and resolved host filters are retained for reproduction.

Metrics, probability maps, and controls. We report AP, AP50, and AP75. Score-ordered same-class one-to-one matching at IoU$\geq$ .5 defines correctness for 15-bin ECE and Brier score. We also report 25-bin class-balanced LaECE_0, threshold sweeps from 0 to .5, and support-fitted temperature, positive-slope Platt, and isotonic baselines. The auxiliary final-score map fits one scalar logit temperature over 2,001 log-spaced values in $[.05, 10]$; being monotone, it cannot change AP. All new experiments use explicit Boolean score-band masks. The legacy-prefix audit, complete parameter grids, metric definitions, and threshold checks appear in the corresponding sections below.

1. Continuous Guard Comparison

The canonical PTPC-B result uses the continuous multiplier defined in the main paper's Guard equation. Its temperature is selected on the opposite source benchmark ($\eta = .04$ for VOC and .01 for COCO) and then frozen across both detector families, every split, and every shot. No target query label selects a parameter. We reuse the archived binary-limit outputs as unchanged paired controls; no binary experiment is rerun. All 42 continuous detector–shot cells are checked against the archived configurations and results. Each of these cells also improves over its paired detector-only baseline; comparisons with the binary-limit control are reported separately below.

Because each displayed binary VOC value represents a $\pm.05$-point rounding interval, we do not infer exact deltas from rounded cells. The canonical mean exceeds the displayed binary mean by 0.672 AP50 for DeFRCN (conservative interval $[0.622, 0.722]$) and 0.596 for MFDC ($[0.546, 0.646]$). DeFRCN has 11 rounding-robust wins, three losses, and one indeterminate cell; MFDC has nine wins and six losses. The continuous rule improves the mean for both detectors, although it does not improve every cell.

Canonical Guard AP and AP50 exceed the binary-limit control in all 6/6 COCO shots for both detectors. Mean AP50 differences are +0.154 for DeFRCN and +0.142 for MFDC. Together with Table 1, this is the direct evidence for making the continuous rule canonical. The PTPC-L rows preserve the encoder-scale controls moved out of the main comparison tables; because no continuous PTPC-L matrix was used for selection, they do not enter the canonical rule comparison.

2. Paired Image-Bootstrap Uncertainty of the Continuous Guard

Evaluation population and measured effect. We evaluate all 21 FSOD-VFM and 17 No-Time target cells from the original cross-dataset-frozen protocol. Parameters are taken directly from the archived opposite-benchmark selection, including $\eta = .04/.01$ for FSOD-VFM on VOC/COCO and $\eta = .04/.005$ for No Time. We do not repeat source selection or use query labels to choose parameters. The effect is continuous-Guard minus native AP50 on VOC and AP on COCO, in percentage points. The benchmark summary is the original equal-weight mean over that host's shot/split cells.

Paired cluster resampling. The sampling unit is a query image, including images with no target-class ground truth. For each of 2,000 replicates, sample the original number of images with replacement and evaluate both predictions on the same multiset. Every copy of a sampled image has its own detections and annotations. A single draw is reused across all shots and splits in a host/benchmark; we average their paired metric differences within the replicate, then take the 2.5th and 97.5th percentiles. This retains image-induced dependence across nested supports and overlapping query sets. NumPy's `default_rng` uses seed `20261020`; ordered query-image IDs must agree across the cells combined in a benchmark mean. We do not bootstrap already-aggregated AP numbers or average per-image AP.

Reconstruction and evaluation checks. We use the archived native predictions, diagnostics, and annotations to reconstruct the original float32 continuous-Guard rule from each region's native score and base/novel similarities, including the original host-specific arithmetic. No-Time scores that are non-finite in both scoring variants are mapped to zero for metrics, as in the original experiments; raw files remain unchanged. Both native and continuous-Guard point metrics must agree with their archived values to absolute tolerance 10^{-8}. COCO-format matching, target categories, crowd/ignore handling, 101-point interpolation, and the 100-detection evaluation cap reproduce the original native-host evaluator on both benchmarks. Per-image matches are cached separately for each score ordering. The resampled accumulator is checked against the official COCO implementation both on the original population and on a duplicate-image resample, to tolerance 10^{-12}. Confidence ties retain stable order within each sampled population.

Interpretation. These are marginal percentile intervals for each stated effect, conditional on the fixed checkpoints, supports, prototypes, and selected rule. They are not simultaneous intervals across all 38 cells. We make no claim of multiplicity-adjusted cellwise significance. They quantify query-image uncertainty of the canonical native-host Guard and do not replace repeated continuous-rule training/support experiments. The binary-limit repeat/support evidence is reported in the following section.

Table 1. Full PTPC PASCAL VOC rule and encoder-scale ablation, novel AP50. The binary-limit results were archived to one decimal; continuous values are shown at their measured precision. Every row uses identical detector outputs within a detector.

Detector	Variant	Split 1					Split 2					Split 3					Mean
		1	2	3	5	10	1	2	3	5	10	1	2	3	5	10	
DeFRCN	PTPC-B, binary limit (displayed)	63.6	68.3	70.3	73.2	71.5	36.4	43.4	58.1	59.8	60.2	55.8	58.3	61.6	65.2	65.6	60.75
	PTPC-B (canonical)	64.69	69.68	70.49	73.58	71.77	38.00	46.41	57.99	60.56	60.08	56.28	58.65	61.63	65.13	66.43	61.43
	PTPC-L, binary limit (displayed)	62.8	68.6	70.0	73.3	71.8	35.3	43.6	57.0	59.6	59.3	54.1	57.5	60.9	64.9	65.6	60.3
MFDC	PTPC-B, binary limit (displayed)	68.7	71.1	71.1	74.8	73.6	43.2	41.5	51.3	60.3	59.9	59.9	64.9	64.8	67.6	67.8	62.70
	PTPC-B (canonical)	70.15	67.37	73.57	75.91	76.40	42.02	43.60	55.98	60.10	60.47	60.78	63.29	64.23	67.21	68.36	63.30
	PTPC-L, binary limit (displayed)	69.1	68.4	73.0	76.2	76.9	40.1	43.3	55.4	58.5	58.9	56.8	64.4	64.2	67.2	67.4	62.7

Table 2. Full PTPC MS COCO rule and encoder-scale ablation, novel AP. PTPC-B binary values are parsed from saved patch-only logs and continuous values from paired frozen runs; the archived PTPC-L rows retain their displayed one-decimal precision. No binary run is reproduced.

Detector	Variant	1	2	3	5	10	30	Mean
DeFRCN	PTPC-B, binary limit	10.2871	15.6605	17.2564	18.4965	21.1024	25.0662	17.9782
	PTPC-B (canonical)	10.5146	15.7153	17.3516	18.5541	21.1142	25.1086	18.0597
	Canonical minus binary	+.2275	+.0548	+.0952	+.0576	+.0118	+.0424	+.0816
	PTPC-L, binary limit (displayed)	9.8	15.0	16.8	18.2	20.8	24.8	17.6
MFDC	PTPC-B, binary limit	12.3051	16.9612	17.5397	19.4950	21.6471	24.5772	18.7542
	PTPC-B (canonical)	12.5195	17.0333	17.6516	19.5652	21.6889	24.6055	18.8440
	Canonical minus binary	+.2144	+.0721	+.1119	+.0702	+.0418	+.0283	+.0898
	PTPC-L, binary limit (displayed)	11.8	16.2	17.1	19.3	21.2	24.5	18.4

Table 3. **Canonical continuous-Guard paired intervals: VOC AP50.** Each entry is continuous Guard minus native, in percentage points, with a marginal 95% paired image-bootstrap interval (2,000 replicates). All configurations are the original cross-dataset-frozen target runs.

Split	Shots	FSOD-VFM gain	FSOD-VFM 95% interval	No Time gain	No Time 95% interval
1	1	+3.56	[2.96, 4.14]	+1.92	[1.16, 2.76]
1	2	+1.64	[1.21, 2.13]	+2.59	[1.95, 3.41]
1	3	+1.34	[0.95, 1.88]	+2.94	[2.18, 3.81]
1	5	+1.69	[1.26, 2.15]	+3.34	[2.64, 4.15]
1	10	+1.81	[1.35, 2.33]	+2.84	[2.25, 3.55]
2	1	+2.19	[1.70, 2.69]	+0.35	[-0.19, 0.97]
2	2	+4.00	[3.28, 4.82]	+0.84	[0.44, 1.27]
2	3	+3.30	[2.69, 3.94]	+1.15	[0.60, 1.72]
2	5	+2.49	[1.97, 3.10]	+1.24	[0.80, 1.73]
2	10	+1.73	[1.26, 2.29]	+1.06	[0.69, 1.47]
3	1	+1.71	[1.20, 2.40]	+1.29	[0.58, 2.14]
3	2	+1.00	[0.62, 1.45]	+1.72	[1.15, 2.31]
3	3	+0.72	[0.32, 1.23]	+0.64	[0.24, 1.00]
3	5	+0.78	[0.41, 1.24]	+1.01	[0.56, 1.48]
3	10	+0.65	[0.29, 1.06]	+0.78	[0.47, 1.14]

3. Run-to-Run Variance

The repeated-fine-tune and shot-specific support-draw results appear in Tables 3 and 4 of the main paper, respectively. This section gives the experimental protocols and additional analysis. All repeated experiments here use the archived *binary-limit* rule. They quantify robustness of that variant; the paired query-image uncertainty of the canonical continuous Guard is reported separately.

Repeated fine-tunes. Main-paper Table 3 lists every one of the fifteen split×repeat cells behind the main paper's first variance study. We run five DeFRCN 5-shot fine-tunes per split from independent optimizer seeds (`cfg.SEED=-1`) on the canonical seed0 support annotations, and recalibrate each with binary-limit PTPC-B on *its own* checkpoint. These archived robustness experiments complement the canonical continuous seed0 matrix; they are not continuous-Guard reruns. The pairing is exact: within a repeat, the detector-only and PTPC rows share the trained weights, the post-NMS boxes, the predicted labels and the support set, and differ only in inference-time scores. All 15/15 paired gains are positive, and the smallest single gain

Table 4. **Canonical continuous-Guard paired intervals: COCO AP.** Each entry is continuous Guard minus native, in percentage points, with a marginal 95% paired image-bootstrap interval (2,000 replicates). All configurations are the original cross-dataset-frozen target runs. No Time has COCO target results at 10/30 shots only.

Split	Shots	FSOD-VFM gain	FSOD-VFM 95% interval	No Time gain	No Time 95% interval
—	1	+0.10	[-0.20, 0.36]	—	—
—	2	+0.22	[-0.06, 0.52]	—	—
—	3	+0.72	[0.51, 0.97]	—	—
—	5	+0.96	[0.75, 1.19]	—	—
—	10	+0.87	[0.70, 1.05]	+0.80	[0.64, 0.98]
—	30	+0.75	[0.60, 0.91]	+0.70	[0.54, 0.88]

(+9.39, Split 2 repeat 1) is more than twenty times the largest baseline standard deviation.

Alternative support draws. Main-paper Table 4 reports the second study: the complete 1/2/3/5/10-shot VOC grid re-run under four alternative *official* support draws, for both detector families. Each cell is the mean±SD of four paired ΔAP50 values. All 60 DeFRCN and all 60 MFDC paired cells are positive. The largest per-cell standard deviations (±3.9 at DeFRCN Split 3 1-shot, ±3.2 at MFDC Split 2 3-shot) occur where the support draw matters most, i.e. at low shot counts. Even in those two cells, the mean gain is respectively 3.5 and 2.3 times the standard deviation, and every underlying paired gain remains positive.

MS COCO. Re-fine-tuning MFDC at 10-shot under the same four support draws gives novel AP of 22.25/22.41/22.34/22.25 with binary-limit PTPC-B against the corresponding detector-only runs, a paired $+2.5_{\pm 0.4}$ AP with 4/4 positive.

4. Fixed-Output Ceiling on MS COCO

Construction. For each novel class c and each COCO IoU threshold $t \in \{0.50, 0.55, \ldots, 0.95\}$ we build the bipartite graph between that class's detections and its ground-truth instances, with an edge whenever IoU $\geq t$, and compute a maximum-cardinality one-to-one matching. If $M_{c,t}$ of N_c instances can be matched, then *no* assignment of scores can produce more than $M_{c,t}/N_c$ recall at that threshold, because COCO evaluation itself matches one detection per ground truth. The exact 101-point interpolated-AP envelope is the fraction of recall levels in $\{0, 0.01, \ldots, 1\}$ not exceeding $M_{c,t}/N_c$; averaging over the 20 novel classes and 10 thresholds gives a rigorous upper bound on the main paper's COCO AP for any method that preserves the boxes and the labels and changes only the confidence. The bound is computed on the exact prediction dumps used for the main paper's numbers.

Table 5 shows that GT-IoU ordering uses one score assignment across all IoU thresholds. Its AP is within 0.01–0.12 of the upper bound in every cell, leaving little room to tighten the bound for these output sets. The gap between PTPC and its bound shows that better ranking could improve AP substantially. The 2025–26 methods exceed the bound, so matching their AP also requires new boxes or labels. Because GT-IoU ordering uses ground-truth boxes, it cannot be used at inference time. Its gain measures possible improvements in ranking without identifying the cause of either performance gap.

Table 5. Complete fixed-output ceiling analysis on MS COCO novel classes. "GT-IoU ordering" scores each detection by its IoU with a same-class ground-truth box; "bound" is the exact label-preserving maximum; "labels ignored" additionally allows relabeling and is reported only to show how much of the bound is set by the labels.

Shots	1	2	3	5	10	30
MFDC + PTPC-B						
Reported AP (canonical)	12.52	17.03	17.65	19.57	21.69	24.61
GT-IoU ordering	21.54	27.09	29.75	33.49	36.47	40.24
Fixed-label bound	21.55	27.11	29.80	33.55	36.56	40.35
Labels ignored	28.80	33.23	36.68	39.07	42.00	45.46
DeFRCN + PTPC-L						
Reported AP (binary-limit control)	9.78	14.99	16.75	18.21	20.80	24.80
GT-IoU ordering	19.40	25.14	28.50	30.94	35.07	40.05
Fixed-label bound	19.45	25.18	28.55	31.03	35.18	40.14
Labels ignored	27.83	31.86	35.21	37.66	41.43	45.34
2024 full-pipeline frontier	5.7	11.0	15.7	21.9	34.0	37.0
VisTex-GLIP [2]	47.9	51.8	52.2	52.6	52.7	53.6
FSOD-VFM [1]	–	–	–	–	44.0	45.8

Instance-level recall ceiling. Table 6 gives the score-independent maximum-matching recall at IoU ≥ 0.5 over all 19,886 novel ground-truth instances. This is a recall ceiling on the post-NMS output set. It is an instance-level (micro) quantity, whereas COCO AP is a class-mean (macro) quantity dominated by different classes, so the two must not be subtracted from one another; nor is either the same as RPN recall.

TIDE cross-check. On the MFDC 10-shot outputs, TIDE novel-class AP50 rises from 33.15 to 37.71 with PTPC. The isolated classification-error contribution falls from 4.98 to

Table 6. Maximum-matching recall (%) at IoU ≥ 0.5 on the fixed post-NMS outputs, class-agnostic / label-preserving.

Shots	1	2	3	5	10	30
MFDC	18.0/11.8	22.0/17.2	27.1/21.6	30.3/26.8	35.6/32.2	38.5/34.8
DeFRCN	23.3/15.2	30.5/24.2	41.0/35.1	48.9/43.8	54.7/49.7	56.5/50.7

1.86 dAP, background error from 3.34 to 2.62, and the false-positive special error from 24.03 to 19.47. The direction agrees with the guard-selectivity diagnostic in the main paper: most of the improvement is removal of false positives and of confidently mislabeled detections.

5. Guard Transfer to Native Foundation-Model Detectors

Why a Guard can help when DINOv2 is already present. FSOD-VFM and No Time to Train already obtain strong region representations from DINOv2. The Guard reuses these features to add a separate base-class comparison. A pipeline can produce strong positive evidence for a novel label without explicitly testing whether a cached base class explains the same region better. We reuse the pipeline's own final-region descriptor, compute δ^+ and δ^- in that native space, and apply the Guard rule after native post-processing. Proposals, boxes, labels, detection counts, native graph diffusion or classification, and NMS are unchanged.

Native-host parameter regimes and label accounting. The fixed CNN setting is not assumed to transfer to native ViT hosts, whose feature similarities and score distributions differ. We therefore distinguish two regimes throughout the paper.

Cross-dataset frozen (strict target-task label budget). For FSOD-VFM, every VOC split uses the single tuple selected on COCO train2017, while every COCO shot uses the single tuple selected on VOC Split 1 trainval. No Time uses the same direction of transfer with its own host-specific source selections. The target task contributes only its normal base/novel supports for prototype construction; no target-task novel label selects or revises a score parameter. Both complete rescoring matrices are computed from saved native scores and similarities on the original detection populations. Table 7 contrasts the canonical continuous Guard with its binary-limit control. For the binary-limit FSOD-VFM control, VOC AP50 improves in 15/15 cells and COCO AP in 5/6, with a -0.045-point change at 1-shot; AP50, ECE, and Brier improve in all 21 cells. The continuous Guard's improvement counts are reported below.

The Guard uses the same temperature grid, specified in advance: $\eta \in \{.0025, .005, .01, .02, .04, .08\}$. The same source validation population that selected the binary tuple chooses η by AP50, breaking ties by AP, then lower Brier score, then lower temperature. For FSOD-VFM, COCO chooses $.04$ for all VOC targets and VOC Split 1 chooses $.01$ for all COCO targets; for No Time the corresponding choices are $.04$ and $.005$. No target label enters either choice. Relative to native, the FSOD-VFM Guard improves AP, AP50, ECE, and Brier in all 21 cells. Relative to the binary limit, it wins 14/21 AP and 18/21 AP50 comparisons. The corresponding No-Time counts are 17/17 against native, 13/17 AP and 14/17 AP50 against the binary limit.

Table 7. Cross-dataset-frozen FSOD-VFM Guard and binary-limit ablation. Parameters are selected on the other benchmark, so target-task novel labels occur only in the K supports. Metrics are means over 15 VOC or 6 COCO cells; AP columns are percentages.

Target (source)	Rule and frozen parameters	AP	AP50	AP75	ECE↓	Brier↓
VOC (COCO)	Native	60.95	78.23	66.45	.1502	.05230
	Binary: $(.01, .85, (.05, 1])$	61.98	79.85	67.54	**.1294**	**.03925**
	Guard: shared tuple, $\eta = .04$	**62.13**	**80.14**	**67.71**	.1307	.04003
COCO (VOC S1)	Native	39.79	53.38	43.20	.2009	.09539
	Binary: $(.05, .50, (.05, 1])$	40.29	54.06	43.76	.1550	.06809
	Guard: shared tuple, $\eta = .01$	**40.40**	**54.23**	**43.88**	**.1543**	**.06660**

Auxiliary-label diagnostic (target-task supervised). To measure host-specific tuning when extra labelled data are available, for each VOC split we construct a deterministic 1,000-image subset of VOC 2007/2012 trainval by SHA-256 ranking, then remove every image used to build that host's 5-shot base or novel prototypes. Numeric IDs and file stems are checked independently; both give zero overlap. The same selected images are materialized in COCO-JSON and VOC-XML layouts where required. We analogously build a 1,000-image COCO train2017 subset disjoint from the 10-shot support images and from minival query test. These subsets are test-disjoint, but their novel annotations are additional target-task labels beyond the K-shot budget.

For every host/diagnostic task we sweep nine nonnegative margins $m \in \{0, .01, .025, .05, .075, .10, .15, .20, .30\}$, six suppression factors $\gamma \in \{.10, .30, .50, .70, .85, .95\}$, and 12 score bands specified in advance, $(\ell_G, u_G]$, for 648 settings. COCO-format selection ranks every tuple by AP50, AP, AP75, then lower fire rate; reevaluates the leading 40 on three deterministic image folds; requires full-set AP and AP75 losses no larger than 0.10 and 0.20 percentage point and worst-fold AP50 loss no larger than 0.25 point; and chooses by mean-fold AP50 gain, worst-fold gain, full AP50 gain, then lower fire rate. One tuple is frozen per host/VOC split and reused at 1/2/3/5/10 shots; one tuple is frozen per host on COCO and reused across every reported COCO shot. This avoids query-test adaptation but does not make the diagnostic strict K-shot.

The auxiliary-label selected values vary across tasks even though both hosts use DINOv2, supporting the use of

Table 8. Auxiliary-label native-host binary-limit diagnostic settings. The 1,000 labelled images are support- and test-disjoint but exceed the strict K-shot label budget. The last column is binary-limit Guard minus native: AP50 for VOC and AP for COCO. Each setting is reused at every query shot in that row's split/dataset.

Host	Validation task	Frozen $(m, \gamma, (\ell, u])$	Δmetric
FSOD-VFM	VOC S1	$(.05, .50, (.05, 1])$	$+2.32$
	VOC S2	$(.025, .50, (.05, 1])$	$+1.50$
	VOC S3	$(.025, .85, (.05, 1])$	$+0.88$
	COCO	$(.01, .85, (.05, 1])$	$+0.32$
No Time	VOC S1	$(.01, .50, (0, .70])$	$+4.32$
	VOC S2	$(.075, .10, (.05, .70])$	$+0.98$
	VOC S3	$(.05, .10, (.05, .70])$	$+1.47$
	COCO	$(.01, .70, (0, .70])$	$+0.40$

host-specific settings.

The saved auxiliary-label sweeps show how performance varies across settings. Across the three VOC splits, the AP50 gap from the best to the 25th retained candidate is only 0.25–0.35 point for FSOD-VFM and 0.07–0.08 for No Time; on COCO it is 0.11 and 0.12 point. Every selected fold change is positive, and even the weakest fold gains 0.64/0.28 VOC AP50 point for FSOD-VFM/No Time. Similar gains occur across several margins, suppression factors, and score bands.

FSOD-VFM reuses its mask-pooled descriptors after graph diffusion and top-100 selection; No Time reuses its mask-pooled DINOv2 ViT-L/14 descriptors. Thirty-four No-Time scores are non-finite in both variants; for metrics only, both paired values are mapped to zero.

Fixed output populations. Rescoring begins after each host's native filtering, and therefore cannot recover a prediction removed upstream. DeFRCN uses score threshold .05, NMS .5, and at most 100 detections per image. FSOD-VFM uses its released minimum score .01, 30 diffusion steps, and top 100 predictions per image. No Time to Train uses SAM2 proposal IoU threshold .4, NMS .5, and at most 100 output instances; it has no additional PTPC prefilter. The resolved configuration is retained for each setting.

Controlled query-time benchmark. Support descriptors and class means are computed once before query evaluation; individual support descriptors are discarded after averaging. The novel module extracts $K|C_n|$ patch descriptors. A distinct-space Guard extracts another $K|C_n| + \sum_k M_k$ support/base descriptors, whereas a patch-RoI Guard can share novel supports. For N_N eligible novel-module detections, work after one full-image encoder pass is $O(d \sum_i |\Omega(b_i)| + N_N|C_n|d)$. A crop-CLS Guard additionally encodes at most N_G crops and performs $O(N_G(|C_n| + |C_b|)d)$ cosine work. Native Guard-only transfer performs only these matrix products on host descriptors. A patch-RoI implementation could reuse the novel module's patch grid; the released ablation path performs a second full-image pass, so we do not claim that unmeasured speedup.

Table 9 times the three inference paths in one sequential job on an NVIDIA GeForce RTX 3090 (driver 580.159.03). Every path uses the same DeFRCN VOC Split 1 5-shot seed0 checkpoint, support set, 4,952-image VOC 2007 test set, data loader, and output limit. The resolved YAML files differ only in output directory and the expected orchestration flags: the raw path disables all post-hoc processing, the contextual path enables PTPC with the Guard off, and the full path changes only the Guard enable flag. The full path uses the deployed patch-only contextual score, crop-CLS Guard, Boolean score bands, and $M = 20$. Its evaluator output matches the canonical $M = 20$ result exactly.

We report Detectron2's complete query-inference timer and pure GPU-compute timer after its standard warm-up. These exclude process/model loading and the one-time offline support-prototype construction. The full query path costs 112.4 ms/image, or 8.90 images/s, and adds 63.9 ms/image over raw detector inference. This is one complete-test-set measurement rather than repeated-run latency variance. The deployed full VOC cache has 5 novel patch prototypes, 5 novel Guard prototypes, and 15 base Guard prototypes: 25 float32 vectors of dimension 768, or 75 KiB. This cache figure excludes the frozen encoder's weights and activations.

Table 9. Controlled query latency on one RTX 3090 over all 4,952 VOC test images. "Total" is Detectron2 inference after warm-up; Δ is total latency above the raw detector. Offline prototype construction is excluded.

Query path	Total (ms/img)	Compute (ms/img)	img/s	Δ (ms/img)
Raw DeFRCN detector	48.514	47.755	20.613	—
Contextual novel module	78.431	77.649	12.750	+29.917
Full PTPC	112.391	111.500	8.898	+63.877

Full native-detector matrices. Table 10 and Table 11 give every strict cross-dataset-frozen FSOD-VFM cell.

Cross-dataset-frozen No-Time-to-Train. The native captures already store every baseline score and maximum base/novel similarity, so changing the source-frozen tuple or η requires no SAM2 or DINOv2 rerun. COCO source validation selects the complete tuple and $\eta = .04$ for all 15 VOC targets; VOC Split 1 selects the complete tuple and $\eta = .005$ for both COCO targets. The Guard raises mean VOC AP50 from 68.07 to 69.65 and mean COCO AP from 36.67 to 37.42, improving AP, AP50, ECE, and Brier over native in every one of the 17 cells. Relative to the source-frozen binary limit it adds 0.39 VOC AP50 and 0.04 COCO AP. More importantly than the mean, graded

Table 10. Cross-dataset-frozen FSOD-VFM PASCAL VOC matrix, novel AP50. The source-selected non-temperature tuple and Guard $\eta = .04$ are fixed across all 15 target cells.

Scoring	Novel Split 1					Novel Split 2					Novel Split 3					Mean
	1	2	3	5	10	1	2	3	5	10	1	2	3	5	10	
Native	77.95	82.98	84.05	86.63	86.47	65.44	69.22	78.54	79.81	82.23	65.54	75.49	79.73	79.02	80.31	78.23
Binary-limit Guard	81.21	84.52	85.22	88.29	88.33	66.94	71.71	81.59	82.11	83.77	66.56	76.18	80.36	79.94	81.08	79.85
Guard	81.51	84.62	85.39	88.32	88.28	67.64	73.23	81.84	82.29	83.96	67.25	76.49	80.45	79.80	80.97	80.14
Guard − binary	+0.30	+0.10	+0.18	+0.03	-0.05	+0.70	+1.52	+0.25	+0.19	+0.19	+0.69	+0.32	+0.09	-0.15	-0.12	+0.28

Table 11. Cross-dataset-frozen FSOD-VFM MS COCO matrix, novel AP. The source-selected non-temperature tuple and Guard $\eta = .01$ are fixed across all target shots.

Scoring	1	2	3	5	10	30	Mean
Native	26.98	35.02	40.37	43.59	45.55	47.26	39.79
Binary-limit Guard	26.93	35.23	40.97	44.44	46.28	47.88	40.29
Guard	27.07	35.23	41.10	44.55	46.41	48.01	40.40
Guard − binary	+0.14	+0.00	+0.13	+0.11	+0.13	+0.13	+0.11

suppression turns the binary limit's one VOC AP50 regression into a gain, so all 15 VOC cells improve. On VOC the binary limit retains slightly lower mean ECE/Brier; both rules nevertheless improve both calibration metrics in every cell. Table 13 reports the complete strict and auxiliary VOC matrices.

Tables 12 and 13 give the auxiliary-label diagnostics over every shot and split. FSOD-VFM retains its archived binary-limit control; the No-Time table also includes its newly evaluated Guard and the strict cross-dataset block. The auxiliary-label No-Time-to-Train block uses the same 1,000-image development budget as the FSOD-VFM diagnostic. We deliberately do not mix published native values with our Guard values: every comparison uses the same official run, support, boxes, labels, and output order. The archived binary limit improves 14/15 FSOD-VFM cells and 13/15 auxiliary-label No-Time cells.

The corresponding MS COCO diagnostic is Table 14. FSOD-VFM's released protocol provides 10/30-shot supports; its 1/2/3/5-shot cells are deterministic nested prefixes of the released 30-shot support and are labelled as extensions. No Time to Train is evaluated at its reported 10/30-shot protocol. The auxiliary blocks use one setting selected on each host's labelled 1,000-image, support- and test-disjoint COCO subset and then freeze it across COCO shots. The additional No-Time cross-dataset block uses only the VOC-Split-1-selected tuple and $\eta = .005$. Query-test labels never select a parameter; only the auxiliary blocks exceed the strict K-shot label budget.

Saved predictions and failure analysis. For each native output we store its native score, maximum base and novel similarities, label, and box. These stored values let us vary guard margin m, suppression γ, and lower or upper score-band limits without rerunning the detector or foundation model. Changing the region descriptor, prototype construction, or support set still requires feature extraction.

For each query run, we rematch the native and binary-limit Guard rankings and examine every suppression. We distinguish useful FP suppression from harmful TP suppression, count crossings at the 0.5 deployment threshold, and assign each fired detection to base-class confusion, novel-class confusion, duplicate, poor-localization, background/weak-overlap, or true-positive-suppression categories. An overall AP gain can hide rare novel true positives that look more like a base prototype. The native-model results are reported below.

In this setting, both native hosts suppress orders of magnitude more false positives than true positives, while the threshold-crossing counts make the associated recall risk explicit. They are descriptive checks rather than substitutes for the paired AP matrices.

6. Frozen-Encoder Sweep

Main-paper Table 5 gives the per-shot Split-1 sweep: all eight frozen encoders improve on the MFDC baseline shot-mean of 64.80. Table 16 extends the DINOv2 family to all three splits under one common novel-module configuration, Table 17 tests the same scale ordering on COCO, and Table 18 reports the prototype-separability diagnostic that explains the DINOv2-G reversal.

Checkpoint provenance. DINOv2 S/B/L/G are the official pretrained `torch.hub` checkpoints. The image–text rows use the exact `timm` weights `vit_base_patch16_clip_224.laion2b_ft_in12k_in1k`, `vit_large_patch14_clip_224.laion2b_ft_in12k_in1k`, `ViT-B-16-SigLIP`, and `ViT-L-16-SigLIP-256`; the latter two carry the `webli` pretrained tag. We use each checkpoint's native input normalization and resolution. For a prefix-free encoder, the guard's crop-global descriptor is the native pre-logit global pool rather than a nonexistent CLS token.

Performance does not increase consistently with model

Table 12. Auxiliary-label FSOD-VFM PASCAL VOC binary-limit diagnostic, novel AP50. Every entry is an exact native/binary-limit pair at that split's tuple; selection uses the additional labelled 1,000-image trainval subset.

Scoring	Novel Split 1					Novel Split 2					Novel Split 3					Mean
	1	2	3	5	10	1	2	3	5	10	1	2	3	5	10	
Native	77.9	83.0	84.0	86.6	86.5	65.4	69.2	78.5	79.8	82.2	65.5	75.5	79.7	79.0	80.3	78.2
+Binary-limit Guard	80.1	84.6	85.5	88.5	88.7	65.1	70.8	81.5	82.3	83.8	67.0	76.4	80.5	80.1	81.2	79.7
Δ	+2.1	+1.6	+1.4	+1.9	+2.2	-0.3	+1.6	+3.0	+2.5	+1.6	+1.5	+0.9	+0.8	+1.1	+0.9	+1.5

Table 13. No-Time-to-Train PASCAL VOC Guard study, novel AP50. The first three-row block transfers the complete COCO-selected tuple and $\eta = .04$; the second uses each split's additional labelled 1,000-image development set and selected $\eta = .005/.04/.0025$. Within each block, **bold**/underlined values are best/second best.

Scoring	Novel Split 1					Novel Split 2					Novel Split 3					Mean
	1	2	3	5	10	1	2	3	5	10	1	2	3	5	10	
Strict: native	71.91	70.76	64.05	69.86	76.55	54.62	62.89	71.63	72.68	74.09	57.70	65.15	68.45	69.30	71.45	68.07
Strict: binary limit	72.43	72.74	66.30	72.96	79.05	54.18	63.04	72.74	73.84	74.97	57.84	66.62	**69.19**	**70.61**	**72.41**	69.26
Strict: Guard	**73.83**	**73.35**	**66.99**	**73.20**	**79.38**	**54.98**	**63.73**	**72.78**	**73.93**	**75.15**	**58.99**	**66.87**	69.08	70.30	72.23	**69.65**
Aux.: native	**71.91**	70.76	64.05	69.86	76.55	**54.62**	62.89	71.63	72.68	74.09	57.70	65.15	68.45	69.30	71.45	68.07
Aux.: binary limit	70.46	72.06	65.65	72.79	78.90	54.42	63.29	72.55	73.70	74.89	**58.82**	66.70	**69.28**	70.44	72.40	69.09
Aux.: Guard	70.56	**72.23**	**65.88**	**72.86**	**79.05**	54.58	**63.40**	**72.78**	**73.99**	**75.18**	58.70	**66.77**	69.26	**70.50**	**72.43**	**69.21**

Table 14. Native foundation-model MS COCO Guard study, novel AP. A dash marks an unsupported shot. FSOD-VFM is an auxiliary-label binary-limit control; No Time additionally gives the strict cross-dataset Guard and its binary limit. Within each host/protocol block, **bold**/ underlined values are best/second best.

Pipeline	Scoring	1	2	3	5	10	30
FSOD-VFM	Native	27.0	35.0	40.4	43.6	45.5	47.3
	+Binary-limit Guard	**27.5**	**35.4**	**41.1**	**44.4**	**46.3**	**47.9**
	Δ	+0.5	+0.4	+0.8	+0.8	+0.7	+0.7
No Time	Cross: native	–	–	–	–	36.29	37.05
	Cross: binary limit	–	–	–	–	37.04	37.71
	Cross: Guard	–	–	–	–	**37.09**	**37.74**
	Aux.: native	–	–	–	–	36.29	37.05
	Aux.: binary limit	–	–	–	–	37.08	37.74
	Aux.: Guard	–	–	–	–	**37.21**	**37.88**

Table 15. Planned native-model failure analysis on VOC Split 1 5-shot. Counts refer to binary-limit-Guard-fired novel detections; net safe is FP minus TP downward crossings at score 0.5.

Pipeline	FP suppressed	TP suppressed	FP↓@.5	TP↓@.5	Net safe@.5
FSOD-VFM	233,900	28	5,774	21	+5,753
No Time to Train	221,622	61	757	14	+743

Table 16. DINOv2 scale sweep on MFDC across all three VOC splits, novel AP50. Each value averages the common four-shot subset $\{1, 2, 5, 10\}$; the last column averages the three split means. All rows share one novel-module configuration, so only the encoder varies.

Encoder	Split 1	Split 2	Split 3	All
DINOv2-S/14	69.09	49.69	62.82	60.53
DINOv2-B/14	71.88	**51.05**	**64.49**	**62.47**
DINOv2-L/14	**72.78**	50.16	63.89	62.27
DINOv2-G/14	69.05	49.00	61.44	59.83

Table 17. DINOv2 scale sweep on MFDC COCO, novel AP. Rows use the common three-shot subset $\{1, 5, 10\}$ and the same novel-module configuration.

Encoder	1-shot	5-shot	10-shot	Mean
DINOv2-S/14	**10.58**	18.11	20.35	16.35
DINOv2-B/14	10.29	**18.50**	**21.10**	**16.63**
DINOv2-L/14	9.78	18.21	20.80	16.26
DINOv2-G/14	8.77	17.05	19.69	15.17

size. L is strongest on Split 1, but B has the best common-shot mean over all splits; G is weakest. This suggests that the separation between region and class descriptors under prototype competition matters more than parameter count alone.

COCO reproduces the all-split VOC ordering: B has the best common-shot mean, while G trails B by 1.46 AP. This cross-dataset agreement rules out the simple explanation that G's reversal comes from one difficult VOC split or support draw.

DINOv2-G's failure is not a lack of signal but a lack of *contrast*: its similarity to the correct prototype is the highest of the four (0.646), yet its similarity to the wrong prototypes rises more than proportionally (0.467), so the margin that competitive scoring exploits collapses to less than half that of B. A softmax over near-uniform similarities carries little information about which novel class the detection

Table 18. Prototype separability on held-out novel ground-truth boxes, aggregated over three splits and five shot settings. *Match* is the percentage of boxes whose nearest class prototype is the correct one; *true*/*wrong* are mean cosine similarities to the correct and to the incorrect prototypes; *margin* is their difference.

Encoder	Match (%)	True cos	Wrong cos↓	Margin
DINOv2-S/14	97.7	0.581	0.205	0.376
DINOv2-B/14	**98.1**	0.558	**0.178**	**0.380**
DINOv2-L/14	97.5	0.575	0.234	0.341
DINOv2-G/14	92.5	0.646	0.467	0.178

belongs to, which is exactly the quantity the competitive-compatibility equation in the main paper feeds into the fusion.

7. Base-Confusability Guard

Selected parameter settings. For the strict cross-dataset-frozen FSOD-VFM control, all VOC cells use the COCO-selected $(m, \gamma, (\ell_G, u_G]) = (.01, .85, (.05, 1])$ tuple and all COCO cells use the VOC-Split-1-selected $(.05, .50, (.05, 1])$ tuple. The strict continuous evaluation additionally freezes the source-selected $\eta = .04$ for VOC and $\eta = .01$ for COCO and verifies that its native/binary rows reproduce the archived binary results exactly. For the separate auxiliary-label diagnostic, Split 1 uses $(m, \gamma) = (.05, .50)$, Split 2 uses $(.025, .50)$, and Split 3 uses $(.025, .85)$, all with $(\ell_G, u_G] = (.05, 1]$. Every analysis uses the applicable source-dataset or split-level setting, without revising parameters on query-test data.

Guard feature space. Table 19 gives all 15 split×shot cells behind the main paper's claim that neither Guard feature space is consistently superior. Holding the novel module at its deployed patch-RoI configuration, we compute the guard descriptor either as the crop-CLS embedding (deployed) or as the novel module's patch-RoI descriptor. The means are 60.75 and 60.79 mean AP50; crop-CLS wins 7 cells and patch-RoI 8, and the largest per-cell difference in either direction is 3.70. The modules have different roles. The novel module must discriminate among similar novel classes, so retaining spatially localized evidence helps. Guard asks a coarser question—whether any base prototype explains the region better than any novel prototype—and both normalized descriptors are ultimately averaged into one vector per class. Their aggregate tie shows that semantic class separation, not spatial detail, dominates this binary comparison. Both cache one d-vector per base class. Patch-RoI removes crop-CLS encoding and permits a shared patch-grid cache, but the released ablation path re-extracts its Guard descriptor; this table is an accuracy comparison, not a measured speed claim. We retain crop-CLS as the Split-1-selected deployed choice, not because it is intrinsically superior.

Table 19. Guard descriptor ablation on DeFRCN VOC, novel AP50. The contextual novel-prototype module is held fixed; only the Guard feature space changes.

Guard descriptor	1	2	3	5	10	Mean
Split 1, crop-CLS	63.58	68.29	70.25	73.17	71.54	69.36
Split 1, patch-RoI	60.97	68.91	70.60	72.87	70.62	68.79
Split 2, crop-CLS	36.42	43.40	58.11	59.80	60.20	51.59
Split 2, patch-RoI	36.93	47.10	58.48	60.13	58.94	52.32
Split 3, crop-CLS	55.83	58.30	61.61	65.19	65.62	61.31
Split 3, patch-RoI	55.99	58.02	61.49	65.52	65.23	61.25
All, crop-CLS	7 of 15 cells best					60.75
All, patch-RoI	8 of 15 cells best					60.79

Novel-positive reference and base-bank cardinality. The canonical Guard uses $\max_{c \in C_n}$ as positive evidence because it asks whether the region belongs anywhere in the novel set; the contextual novel module separately assesses the fixed predicted label. To test whether this distinction drives the result, we replace that maximum by only the predicted-label prototype while changing nothing else. Table 20 shows a -0.100-point three-split AP50 change. Mean AP/AP75 change from 37.911/38.427 to 37.842/38.326. Split 2 moves slightly in the opposite direction, so the max-reference advantage is small rather than universal.

The same table tests the number B of retained base classes at fixed $M = 20$. For $B = 5, 10$, three SHA-256-ranked class orders, specified in advance, create deterministic nested subsets; $B = 15$ is the complete bank and is invariant to ordering. Pooled AP/AP50/AP75 rise from 36.430/63.331/37.098 at $B = 5$, to 36.946/64.407/37.534 at $B = 10$, and 37.911/66.182/38.427 at $B = 15$. Individual subsets are not monotone: the largest order spread is 3.926 AP50 points on Split 2 at $B = 5$, and one such subset is 0.091 point above the full bank on that split. More base classes improve average performance, but the effect also depends on which classes are included. No test value selects an order, cardinality, or reference rule; the complete-bank/max-novel configuration was already fixed.

Per-class gains and regressions. Table 21 lists every novel class, with the guard's effect on AP50 averaged over shots at the deployed $m = 0.05$. The guard helps 10 of 15 entries and hurts 5, for a net $+3.84$ mean AP50. The pattern is coherent rather than random: the largest gains (cow $+23.0$ on Split 2, sheep $+16.4$ on Split 3, cow $+19.3$ on Split 1) and the largest loss (horse -7.6 on Split 2) are all members of the same visually-similar quadruped group,

Table 20. Guard novel-reference and base-class-count controls on DeFRCN+PTPC-B VOC 5-shot seed0, novel AP50. The detector, contextual novel module, $M = 20$, Guard descriptor, margin, suppression, and score bands are fixed. "Mean" averages the three splits.

Control	Order	Split 1	Split 2	Split 3	Mean
Novel positive reference; all 15 base classes					
Max over novel set	—	73.153	60.178	65.216	66.182
Predicted label only	—	72.908	60.261	65.079	66.083
Predicted minus max	—	-0.246	+0.083	-0.137	-0.100
Base-bank cardinality; max over novel set					
$B = 5$	0	69.046	56.344	63.843	63.078
$B = 5$	1	69.974	60.269	60.578	63.607
$B = 5$	2	69.562	59.448	60.915	63.308
$B = 5$, order mean	—	69.527	58.687	61.779	63.331
$B = 10$	0	70.597	59.699	64.818	65.038
$B = 10$	1	70.924	59.902	61.536	64.121
$B = 10$	2	70.834	59.653	61.701	64.063
$B = 10$, order mean	—	70.785	59.752	62.685	64.407
$B = 15$, complete bank	—	73.153	60.178	65.216	66.182

and which way a class moves depends on whether its confusers are in the base set (gain) or whether a true novel instance itself lies closer to the base-class prototypes (loss). VOC's changing split assignment makes this visible: cow on Split 1 and sheep on Split 3 are tested against several base quadrupeds, whereas noisy few-shot horse prototypes on Split 2 allow real horses to be rejected as base-like. The margin recovery below supports this explanation: requiring a larger base-over-novel gap restores horse recall while preserving most cow and sheep gains.

Table 21. All per-class ΔAP50 values (guard on minus guard off), averaged over 1/2/3/5/10 shots at $m = 0.05$, DeFRCN VOC.

Improved (10/15)			Regressed (5/15)		
S2	cow	+23.00	S2	horse	−7.60
S1	cow	+19.31	S2	sofa	−1.46
S3	sheep	+16.44	S2	bottle	−1.37
S3	boat	+2.42	S2	aeroplane	−0.73
S1	bus	+1.64	S1	bird	−0.21
S3	motorbike	+1.53			
S1	motorbike	+1.41			
S1	sofa	+1.36			
S3	sofa	+0.92			
S3	cat	+0.89			
Mean over all classes				+3.84	

The recall cost is a threshold effect, not a deletion. The guard multiplies confidences by γ; it never removes a detection. We verified this empirically rather than by appeal to the definition. Patching the VOC evaluator to emit per-class recall at score threshold 0—the maximum attainable recall—and re-running Split 2 with the guard on and off gives *identical* values to four decimal places in all 25 class×shot comparisons (maximum absolute difference 0.0000; e.g. horse 10-shot 93.68 both ways). Every true positive the guard demotes remains recoverable by reranking.

At an operating threshold the cost is real, and Table 22 shows both the cost and its remedy. At $m = 0.05$, Split-2 horse loses 19.4 points of recall@0.5 averaged over shots, and Split-2 cow loses 61.5 points at 2-shot in the worst single cell. Raising the guard margin to $m = 0.15$ matches no-guard horse recall@0.5 at 3- and 5-shot, comes within 0.6 point at 10-shot, and recovers most of it at 1–2 shot. It also fully restores the 2-shot cow cell (68.4, identical to no-guard) while retaining the large measured cow and sheep AP50 gains (+27.4 and approximately +16 at $m = 0.15$). The price is 0.46 mean AP50 (66.16 at $m = 0.05$ versus 65.70 at $m = 0.15$) as some base false positives return. Increasing the Guard margin recovers recall while keeping the detector's confidence threshold unchanged. We keep the Split-1-selected deployed $m = 0.05$; $m = 0.10$ is only 0.09 mean AP50 higher, while $m = 0.15$ is the recall-sensitive alternative.

The Guard is applied only to novel-labeled outputs. Consequently, the ordered score list for every base class is identical with Guard on and off, so base AP is mathematically invariant for the fixed output set; generalized performance can change only through its novel component.

Table 22. Guard margin as a precision/recall control, Split 2. *Top:* horse recall at score threshold 0.5. *Bottom:* horse AP50. "off" disables the guard.

Shots	1	2	3	5	10
horse recall@0.5					
guard off	78.5	80.5	80.2	87.6	88.8
$m = 0.05$	19.8	57.8	75.3	83.9	81.9
$m = 0.10$	32.2	60.3	79.3	86.2	84.5
$m = 0.15$	54.3	75.3	**80.2**	**87.6**	88.2
horse AP50					
guard off	58.8	67.7	65.2	74.0	70.5
$m = 0.05$	36.8	61.7	61.6	72.0	66.0
$m = 0.10$	37.4	57.4	63.8	73.9	67.4
$m = 0.15$	46.7	64.1	65.3	74.5	70.1

Auxiliary-label continuous-margin comparison. We also report the earlier within-VOC sensitivity study that motivated the strict cross-dataset control above. It is not used for a headline comparison or to select either cross-dataset-frozen temperature. Its selected temperatures are $\eta = .01/.005/.02$ on Splits 1/2/3. The best continuous-margin candidate is $+0.195/-0.068/-0.033$ AP50 relative to the binary-limit Guard on the three labelled 1,000-image diagnostic sets, respectively. The changes are small and vary in direction across splits. These additional target-task labels place the experiment outside the strict K-shot

budget. We freeze the temperatures and report both rules on query test. Table 23 shows a paired $+0.23$ mean AP50 gain, 14/15 wins, and a $+0.33$-point gain over 1–3 shots. The single loss is 0.001 point. Mean AP/AP75 also rise by 0.15/0.15 point. Calibration changes little: ECE changes $0.0985 \rightarrow 0.0989$ and Brier $0.02539 \rightarrow 0.02550$. By size, AP_S changes $19.12 \rightarrow 18.98$, while AP_M and AP_L improve $46.32 \rightarrow 46.47$ and $75.58 \rightarrow 75.85$; the gain is therefore not driven by the smallest boxes. Shot conditions share protocol structure and are not independent statistical replicates, so we report the paired effect and win count rather than an invalid 15-sample significance test.

Table 23. Binary-limit versus auxiliary-label continuous-margin Guard comparison on FSOD-VFM VOC. Entries are novel AP50 on identical outputs; columns are shots. The extra 1,000 labelled images per split are test-disjoint but outside the strict K-shot budget.

Split	Rule	1	2	3	5	10	Mean
1	Binary limit	80.08	84.59	85.49	88.54	88.70	85.48
1	Continuous-margin ($\eta = .01$)	80.94	84.73	85.59	88.66	88.80	85.74
2	Binary limit	65.14	70.80	81.51	82.33	83.84	76.72
2	Continuous-margin ($\eta = .005$)	65.44	71.25	81.97	82.51	83.94	77.02
3	Binary limit	67.02	76.38	80.50	80.07	81.18	77.03
3	Continuous-margin ($\eta = .02$)	67.38	76.57	80.60	80.07	81.18	77.16
All 15 cells		Binary 79.75; continuous-margin 79.98					$+0.23$

Absolute novel-evidence negative control. The main paper distinguishes a relative Guard from a generic low-similarity filter. For each split and each control family, we select the absolute threshold $a \in \{.30, .35, \ldots, .70\}$ on the same additional labelled diagnostic population and freeze it across all five query-test shots. This is likewise a auxiliary-label, non-strict-K-shot control. The selected thresholds on Splits 1/2/3 are $.45/.55/.50$ for binary absolute-only, $.40/.50/.65$ for continuous-margin absolute-only, $.35/.45/.45$ for binary relative-or-absolute, and $.30/.45/.35$ for binary-relative-or-continuous-margin-absolute; the latter two continuous-margin families use $\eta_a = .05$.

Table 24 shows the complete 15-cell means. Absolute-only suppression touches 44.6–52.9% of detections and improves reliability, but its AP50 remains at the native level ($78.19/78.26$ versus 78.23). In contrast, the binary-limit relative Guard reaches 79.75. This is the binary-limit mechanism control summarized in the main paper, rather than its canonical continuous Guard result. Adding auxiliary-label absolute evidence to it changes AP50 by only $+0.004$ point for the binary OR and $+0.019$ point for the continuous-margin OR. This negative result rules out generic absolute-cosine rejection as the source of the Guard's ranking gain.

Table 24. Absolute novel-evidence controls on FSOD-VFM VOC, mean over 15 cells. AP columns are percentages; Touched is the fraction of fixed detections whose score is reduced.

Rule	AP	AP50	AP75	ECE↓	Brier↓	Touched
Native	60.95	78.23	66.45	0.1502	0.05230	0.0%
Absolute only, binary	61.00	78.19	66.55	0.1034	0.02886	44.6%
Absolute only, continuous-margin	61.03	78.26	66.57	0.1136	0.03255	52.9%
Relative Guard, binary limit	61.96	79.75	67.55	0.0985	0.02539	47.8%
Relative $\vee$ absolute, binary	61.97	79.75	67.58	0.0976	0.02503	48.8%
Relative $\vee$ absolute, continuous-margin	61.99	79.76	67.61	**0.0976**	**0.02491**	52.9%

Score-band semantics audit. The mathematical rule selects detections by a Boolean predicate on the score entering each module. A legacy implementation instead counted scores in the band and sliced that many detections from the detector's original ordered prefix. Those operations coincide before any reordering, but need not coincide after the novel module changes scores. We therefore reran one representative canonical DeFRCN+PTPC-B cell (VOC Split 1, 5-shot seed0, $M = 5$) while changing only this eligibility operation. The legacy-prefix and Boolean results are $42.0257/71.7305/43.6274$ and $41.9931/71.7561/43.6236$ AP/AP50/AP75, respectively: deltas of $-0.0326/+0.0256/-0.0038$ point and a maximum absolute change of 0.033 point. This is an implementation audit, not a selected result. It establishes the scale in one exact paired cell rather than claiming an exhaustive rerun of the earlier matrix; all new base-support-budget experiments and the released code use the explicit Boolean rule.

Base-prototype support-budget diagnostic. Table 25 gives every split in the diagnostic described in the main paper. Within each split, resolved configs differ only in M and the output directory. The supports are deterministic nested prefixes of the canonical base-loader ordering; M crops are averaged into one vector per base class, so this changes offline prototype estimation rather than the number of runtime comparisons. The Split 1, $M = 5$ row reuses the exact Boolean audit output above rather than rerunning the same cell.

Split-mean AP50 rises from 65.42 at $M = 5$ to 66.18 at the predeployed $M = 20$, then changes by only $+0.15$ point at $M = 50$; mean AP and AP75 change by less than 0.03 point in that last step. The within-split AP50 spans are 1.40/1.13/0.33 points. We report this as descriptive budget sensitivity: the three VOC splits are category partitions, not independent replicates, and one nested ordering does not establish invariance to resampled base supports. No query-test value selects a row, and the released default remains $M = 20$.

Table 25. DeFRCN+PTPC-B base-prototype support budget on VOC 5-shot seed0. Split columns are novel AP50; mean columns average the three splits. All metrics are percentages.

M	Split 1 AP50	Split 2 AP50	Split 3 AP50	Mean AP	Mean AP50	Mean AP75
5	71.76	59.63	64.88	37.39	65.42	37.92
10	72.19	59.82	65.04	37.57	65.68	38.07
20	73.15	60.18	65.22	37.91	66.18	38.43
50	73.07	60.76	65.17	37.89	66.34	38.43

Monotone calibration of the final Guard score. We fit one scalar logit temperature per split using $\sigma(\mathrm{logit}(\mathrm{clip}(s, 10^{-7}, 1-10^{-7}))/T)$ and Bernoulli NLL on the same additional labelled 1,000-image diagnostic population, separately for the native and final binary-limit-Guard scores. This fit uses auxiliary target-task labels and is not a strict K-shot result. The fitted native/final temperatures are $(.156/.239)$, $(.163/.261)$, and $(.165/.149)$ on Splits 1–3, and are frozen across shots. We use the binary-limit rule here to isolate final-score calibration separately from the continuous-margin comparison. Table 26 gives the complete 15-cell means. Native temperature scaling substantially improves reliability but cannot alter ordering. The Guard supplies the missing semantic reordering, and temperature scaling then calibrates its *final* score. AP, AP50, and AP75 are exactly invariant before versus after each temperature transform in every cell (maximum numerical difference $< 10^{-10}$). Thus the two operations are complementary, not competing explanations for the AP gain.

Table 26. Auxiliary-label final-score temperature calibration on FSOD-VFM VOC, mean over 15 cells. Temperatures use 1,000 additional labelled, test-disjoint images per split; AP columns are percentages.

Score	AP	AP50	AP75	ECE↓	Brier↓	LaECE_0↓
Native	60.95	78.23	66.45	0.1502	0.05230	0.1505
Native + temperature	60.95	78.23	66.45	0.0400	0.01908	0.0405
Binary-limit Guard	61.96	79.75	67.55	0.0985	0.02539	0.0987
Binary-limit Guard + temperature	61.96	79.75	67.55	**0.0117**	**0.00641**	**0.0122**

8. Split-Wise Hyperparameter Sensitivity

Table 27 expands the main paper's hyperparameter study with per-split values. The deployed setting ($\tau = 0.1$, $m = 0.05$, $\gamma = 0.3$) was chosen on Split 1, so Split 2–3 measure transfer rather than tuning. We retain that setting when reporting the three-split mean. The temperature sweep has a broad interior optimum: very low τ makes support noise dominate the prototype softmax, whereas high τ flattens the class competition that is supposed to reorder detections. The margin is stable around 0.05–0.10 because it changes which ambiguous detections are suppressed; this is the main precision–recall control. Fusion weight has the largest effect because it acts on every detection, while γ changes only the magnitude of suppression for detections that meet the Guard condition. Its 0.42-point range is therefore small. The best alternatives to the deployed values improve the mean by only 0.21 AP50 for τ, 0.10 for m, and 0.21 for γ, supporting stability rather than joint optimality.

Table 27. Hyperparameter sensitivity on DeFRCN VOC 5-shot, split-wise novel AP50. Deployed values in bold; all remaining parameters are held at their deployed values in each sweep.

Value	Split 1	Split 2	Split 3	Mean
Temperature τ				
0.05	71.06	60.17	64.72	65.32
0.10	73.14	60.17	65.19	66.16
0.20	73.64	60.13	65.34	66.37
0.50	72.37	59.55	65.02	65.64
1.00	70.48	57.90	63.68	64.02
Guard margin m				
0.000	71.85	60.33	64.40	65.53
0.025	72.62	60.21	64.94	65.92
0.050	73.14	60.17	65.19	66.16
0.100	73.01	60.43	65.34	66.26
0.150	72.20	60.02	64.89	65.70
Fusion weight w_ϕ				
0.20	69.91	58.69	63.86	64.15
0.40	71.52	60.14	65.01	65.56
0.60	73.14	60.17	65.19	66.16
Suppression γ				
0.10	72.83	59.85	65.19	65.96
0.20	73.00	60.00	65.21	66.07
0.30	73.14	60.17	65.19	66.16
0.50	73.24	60.31	65.20	66.25
0.70	73.52	60.43	65.17	66.37

9. Alternatives to PTPC's Scoring Choices

The main paper's design-space ablation turns PTPC's components on and off. A separate question is how each component compares with a simpler alternative. Table 28 compares the two scoring choices. *Uniform pooling* replaces the center-weighted Gaussian pooling in the main paper with an unweighted mean over the identical selected token set $\Omega(b_i)$, so it isolates the weighting scheme from the choice of which tokens to pool. *Temperature scaling* is the standard scalar post-hoc calibrator applied to the detector's own logits, with the temperature fitted on the novel support set. Temperature scaling reaches the best ECE in the table while leaving detector AP50 unchanged because it preserves detection order. Uniform pooling is within 0.03 AP50 and 0.001 ECE of the deployed center-weighted variant, suggesting that token selection contributes more to the result than Gaussian weighting. Once tokens are restricted to the detector box, the scale-adaptive Gaussian is broad enough to change their relative weights without changing which semantic evidence is present. The competitive prototype rule therefore sees nearly the same descriptor under both pool-

ing choices.

Table 28. PTPC's scoring choices against the simpler alternatives, DeFRCN VOC 5-shot, mean over the three splits. All four rows share one checkpoint, support set and protocol, so they are directly comparable; they are not comparable to the factorial design ablation in the main paper, which uses its own matched checkpoint triplet. Within each metric, **bold** and underlined mark the best and second-best values; ties share emphasis.

Scoring	Mean AP50	ECE↓	Brier↓
Detector only	48.78	0.147	0.058
Temperature scaling	48.78	**0.061**	0.057
PTPC, uniform pooling	66.13	0.114	**0.043**
PTPC, center-weighted (deployed)	**66.16**	0.113	**0.043**

10. Detector-Family Transfer and Calibration Scope

Table 29 reports the complete FSCE transfer grid, identified in the main paper as a supplementary detector-family control. Each pair uses the same checkpoint, support examples, boxes, predicted labels, and evaluator; PTPC changes scores only. The gain is positive in all 15 split×shot cells (overall mean $+9.52$ AP50), rather than an aggregate hiding cellwise reversals or a comparison to a differently trained FSCE model.

Table 29. Exactly paired FSCE novel AP50 on VOC. "Base" is the frozen FSCE checkpoint and "+PTPC" rescores its identical detections.

Split	Scoring	1	2	3	5	10	Mean
1	Base	46.10	50.84	53.69	62.85	63.74	55.44
1	+PTPC	**60.88**	**62.06**	**63.75**	**70.73**	**70.11**	**65.51**
2	Base	27.74	31.97	43.03	46.15	50.49	39.88
2	+PTPC	**35.39**	**38.66**	**52.54**	**58.58**	**59.42**	**48.92**
3	Base	38.03	44.33	48.33	55.49	58.40	48.91
3	+PTPC	**53.95**	**56.61**	**56.60**	**61.35**	**63.40**	**58.38**

The 15-cell FSCE result establishes detector-family transfer of *ranking accuracy*; it is not used to claim detector-wide calibration transfer. The ECE and Brier comparison in Table 28 is scoped to the exactly paired DeFRCN study (ECE $0.147 \rightarrow 0.113$, Brier $0.058 \rightarrow 0.043$). Ranking and marginal calibration are therefore kept as separate claims.

11. Paired Failure Analysis

Scope and matching. We compare detector-only and full PTPC on the exact DeFRCN VOC `voc_2007_test_novel2` Split 2 5-shot seed0 output set. Detection identities are fixed. We independently rerun score-ordered, same-class, one-to-one matching at IoU 0.5 for each score assignment, then classify all $40{,}583/40{,}583$ detections whose scalar score changed. A true positive can therefore be rescued or displaced when reordering changes which duplicate claims a ground-truth instance.

Table 30. Outcome taxonomy over every score-changed detection. Percentages use all 40,583 changed detections as denominator.

Paired outcome	Count	Fraction
False positive demoted	29,839	73.53%
False positive promoted	9,320	22.97%
True positive promoted	1,312	3.23%
True positive demoted	96	0.24%
True positive rescued by reordering	8	0.02%
True positive displaced by reordering	8	0.02%

Most changes are beneficial false-positive demotions, but the 9,320 false-positive promotions account for a substantial share of the changes. Because the Guard never increases a score, every promotion originates in the contextual novel-prototype module: a background region or poorly placed box can share class texture with a novel prototype even though it is not a valid object. Prototype compatibility is therefore not a substitute for objectness or localization, and fusion with detector confidence remains necessary. In comparison, 104 original true positives are demoted or displaced. These counts help explain the net AP gain and show the errors that remain despite the overall improvement.

Fixed-output geometry. We next separate errors that scoring can potentially repair from those fixed by geometry or label. A detection is *localizable same-class* if some same-class GT has IoU≥ 0.5; *wrong label* if its nearest GT has IoU≥ 0.5 but a different class; *poor localization* if nearest-GT IoU is in $[0.1, 0.5)$; and *no nearby annotation* below 0.1.

Table 31. Geometry categories for the same changed detections. Score changes cannot correct the localization and label errors in the middle two groups.

Fixed-output geometry	Count	Fraction
Background / no nearby annotation	34,562	85.16%
Poor localization	3,179	7.83%
Wrong predicted label	1,139	2.81%
Localizable same-class object	1,703	4.20%

All 1,424 promoted/demoted/rescued/displaced true-positive events lie in the localizable same-class group. By contrast, the 4,318 poor-box or wrong-label detections remain false regardless of score. The background bucket can include unannotated base objects under novel-only VOC evaluation, so it is not relabeled as pure base leakage.

Operating-threshold crossings. Table 32 reports exactly which predictions cross common deployment thresholds. Downward FP crossings are beneficial and downward TP crossings harmful; upward TP crossings are beneficial and upward FP crossings harmful. At threshold 0.5, aeroplane, cow, and horse have approximately nonnegative net crossing balance, whereas bottle and especially sofa have more harmful promotions. The score itself remains available, so this table supports application-specific threshold or Guard-margin selection. The 2,255 upward FP crossings also explain why a method can improve global AP yet behave worse at one operating point: AP integrates the full ranked list, whereas a deployed threshold is sensitive only to membership near its cutoff.

Table 32. Threshold crossings. TP/FP status is taken from the score assignment on the side to which the detection crosses.

Threshold	Down	TP down	FP down	Up	TP up	FP up
0.05	14,442	0	14,442	0	0	0
0.30	3,311	32	3,279	2,989	304	2,685
0.50	1,243	21	1,222	2,697	442	2,255

12. Guard-Selectivity Diagnostic

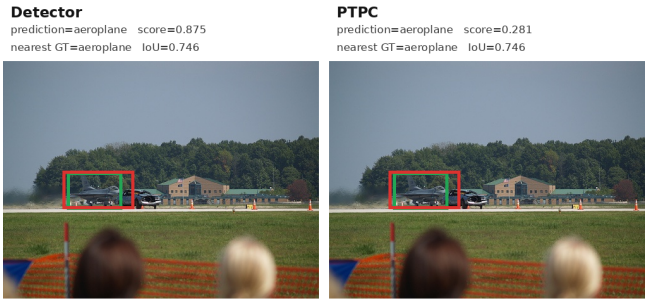
Table 33 gives the counts behind the guard-selectivity rates quoted in the main paper. We log every post-NMS detection whose fused score falls in the guard's intervention band $s \in (0.05, 1.0]$ and bucket it by its maximum IoU with annotated novel ground truth at threshold 0.5. Because the novel-only annotations omit base objects, the no-novel-overlap bucket combines background, base objects and mislocalized novel predictions; it must not be interpreted as pure base leakage. A novel-overlapping detection is a candidate true positive, so its suppression rate directly exposes the guard's recall risk. Both feature spaces are selective, but neither is harmless.

Table 33. Guard selectivity on DeFRCN VOC, aggregated over three splits and all five shots. The novel-module descriptor is patch-RoI; rows vary only the Guard descriptor.

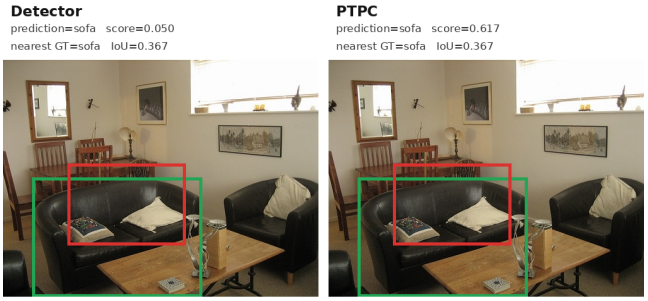
Guard descriptor	Overlap bucket	Checked	Suppressed	Rate
Crop-CLS (deployed)	No novel GT	499,824	399,198	79.9%
Crop-CLS (deployed)	Novel GT	34,113	3,614	10.6%
Patch-RoI	No novel GT	499,824	453,205	90.7%
Patch-RoI	Novel GT	34,113	3,348	9.8%



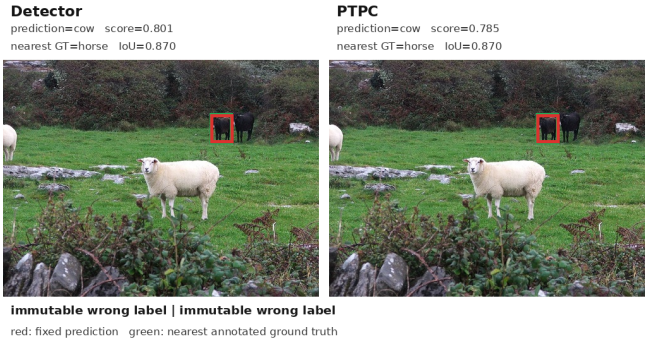
(a) TP demoted: horse, IoU 0.733, 0.912 → 0.279.



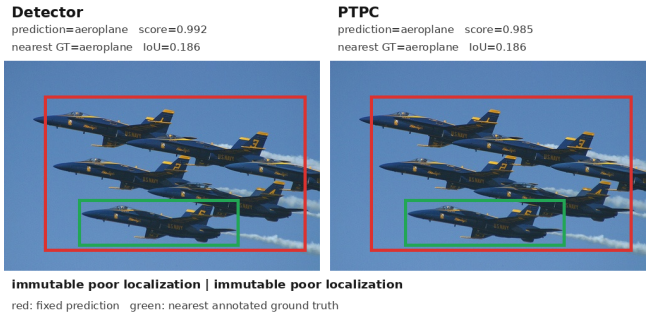
(b) TP displaced: aeroplane, IoU 0.746, 0.875 → 0.281.



(c) FP promoted: sofa, IoU 0.367, 0.050 → 0.617.



(d) Wrong fixed label: cow prediction overlaps horse at IoU 0.870.



(e) Poor fixed box: aeroplane IoU 0.186, 0.992 → 0.985.

Figure 1. **Selected examples of each failure type.** Red denotes the fixed prediction, green the nearest annotated GT. The first three are harmful scoring/reordering events; the last two illustrate errors that scalar rescoring cannot repair.

References

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